

MAINT-AI Synthetic Maintenance Guide

Industrial Air Compressor — AC-300

Document Notice

Synthetic demonstration document created for the MAINT-AI Sprint 2 RAG prototype. It is not a manufacturer-issued service manual.

1. Equipment Overview

Equipment: Reciprocating Industrial Air Compressor

Model: AC-300

Typical application: plant compressed-air systems

Normal operating status: Operational

2. Routine Maintenance

Inspect air filters, check oil level where applicable, inspect hoses and fittings for leaks, monitor discharge temperature, and drain accumulated moisture according to the maintenance schedule. Keep cooling surfaces clean.

3. Fault: COMP-HIGH-05 — High Discharge Temperature

Symptoms: discharge temperature rises above the normal operating range and the compressor may enter thermal protection.

Possible causes include restricted airflow, dirty cooling surfaces, high ambient temperature, low lubricant level where lubrication is applicable, or operation beyond the rated duty cycle.

4. Troubleshooting: COMP-HIGH-05

1. Stop the compressor if the site's high-temperature shutdown threshold is reached.
2. Check cooling airflow and inspect ventilation openings.
3. Clean accessible cooling surfaces after isolation.
4. Check lubricant level if the compressor model uses an oil-lubricated system.
5. Verify that operating demand is within the rated duty cycle.
6. Escalate persistent overheating for qualified inspection.

5. Fault: COMP-VIB-06 — Excessive Vibration

Possible causes include loose mounting, coupling or drive problems, bearing wear, or an unbalanced rotating component. Inspect mounting points and external connections after isolation. Persistent abnormal vibration requires further maintenance assessment.

6. Safety

Depressurize the system before servicing pneumatic components. Isolate electrical power before removing guards or accessing mechanical components. Never bypass thermal protection or other safety controls to keep the compressor running.